ATLAS Radio

SOLID STATE SSB TRANSCEIVERS

ATLAS-210x
10 through 80 meters

ATLAS-215x
15 through 160 meters

THE PERFECT MOBILE RIG.
ONLY 9½ in. wide, 3½ in. high, 9½ in. deep.

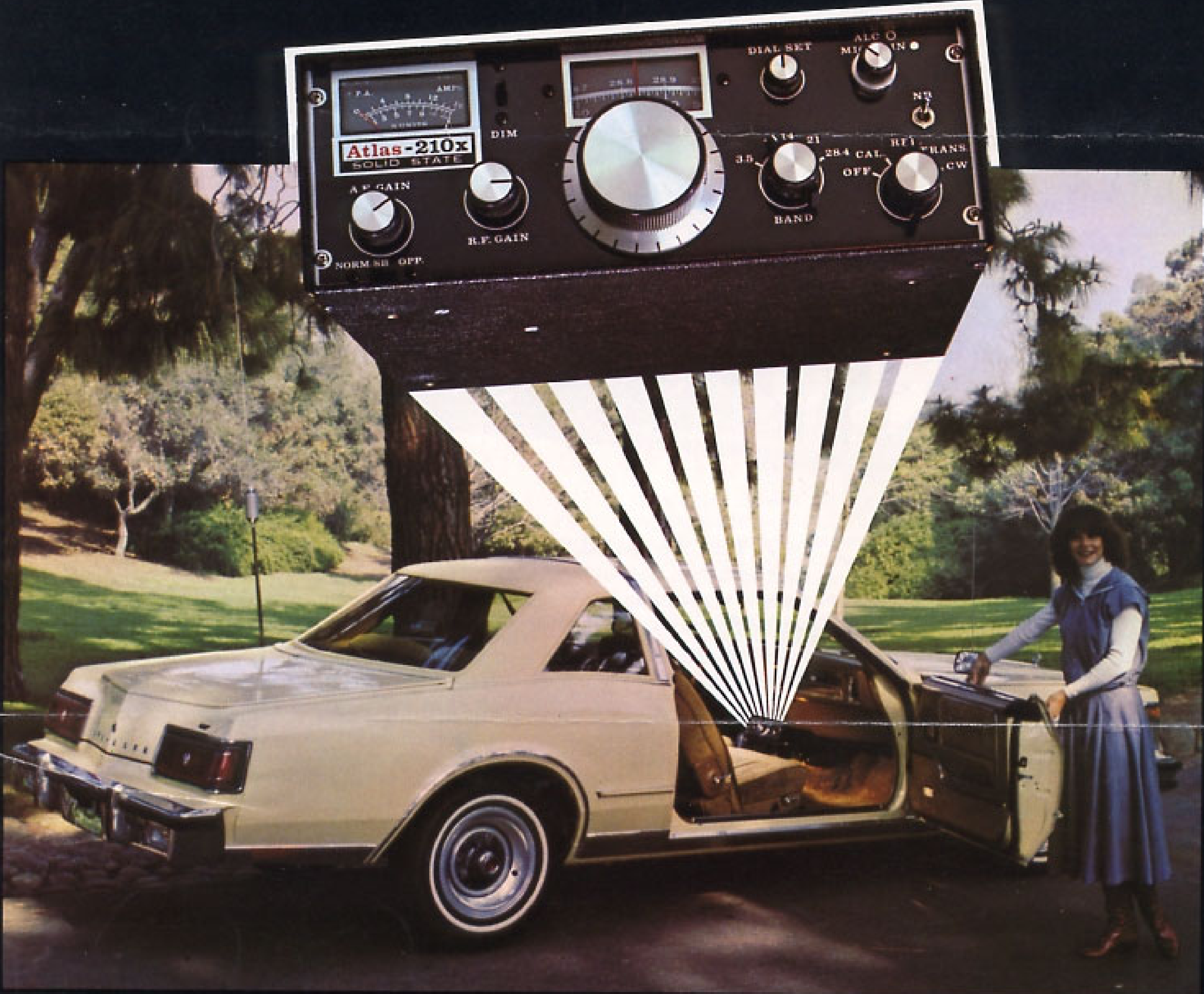
200 WATTS* P.E.P. INPUT   SSB and CW
NO TRANSMITTER TUNING

*120 watts on 10 meters

AMERICAN MADE AND GUARANTEED BY
ATLAS RADIO INC.
92054 Phone (714) 433-1983
Service Direct Line (714) 433-9591
TWX 910-322-1397

ATLAS 210x/215x

5 BAND—200 WATT—ALL SOLID STATE
HF SSB/CW TRANSCEIVER



The Perfect Mobile Rig

ATLAS
RADIO INC.

MADE IN AMERICA

SEVEN POUNDS OF DYNAMITE!

DON'T LET ITS SMALL SIZE FOOL YOU. The Atlas transceiver is packed full of the most advanced, state-of-the-art engineering, and provides unequaled performance in both transmit and receive modes. There is no other transceiver on the market with as many outstanding superior features, regardless of size.

COMPLETELY SOLID STATE DESIGN. 4 I.C's, 18 transistors, and 32 diodes. Years of cool, trouble-free operating pleasure. The final transistors are fully protected against infinite SWR and thermal runaway. Even with constant operating they should never need replacement.

TOTALLY broadbanded. No transmitter tuning or loading controls. No receiver preselector controls. Modern design makes these unnecessary. Instant QSY and band change.

FREQUENCY COVERAGE: 1800-2000, (Model 215x only), 3500-4000 kHz, 7000-7500 kHz, 14,000-14,500 kHz, 21,000-21,500 Khz, 28,400-29,400 kHz, (Model 210x only). The 10 meter band may be easily owner adjusted on the 210x to cover any 1000 kHz portion of the band. Tuning rate is 22 kHz per revolution, with 1 kHz increments on the dial skirt, (2 kHz on 10 meters).

MARS OPERATION: A crystal oscillator is available for fixed channel operation within the amateur bands, as well as outside the regular VFO band edges. For MARS operation, the extended frequency coverage is as follows: 1800-3000 kHz, (215x only), 3000-5200 kHz on 80 meter band, 5800-10000 kHz on 40 meter band, 13,800-14,900 kHz on 20 meter band, 20,600-21,600 kHz on 15 meter band, 28,000-30,000 kHz, (210x only).

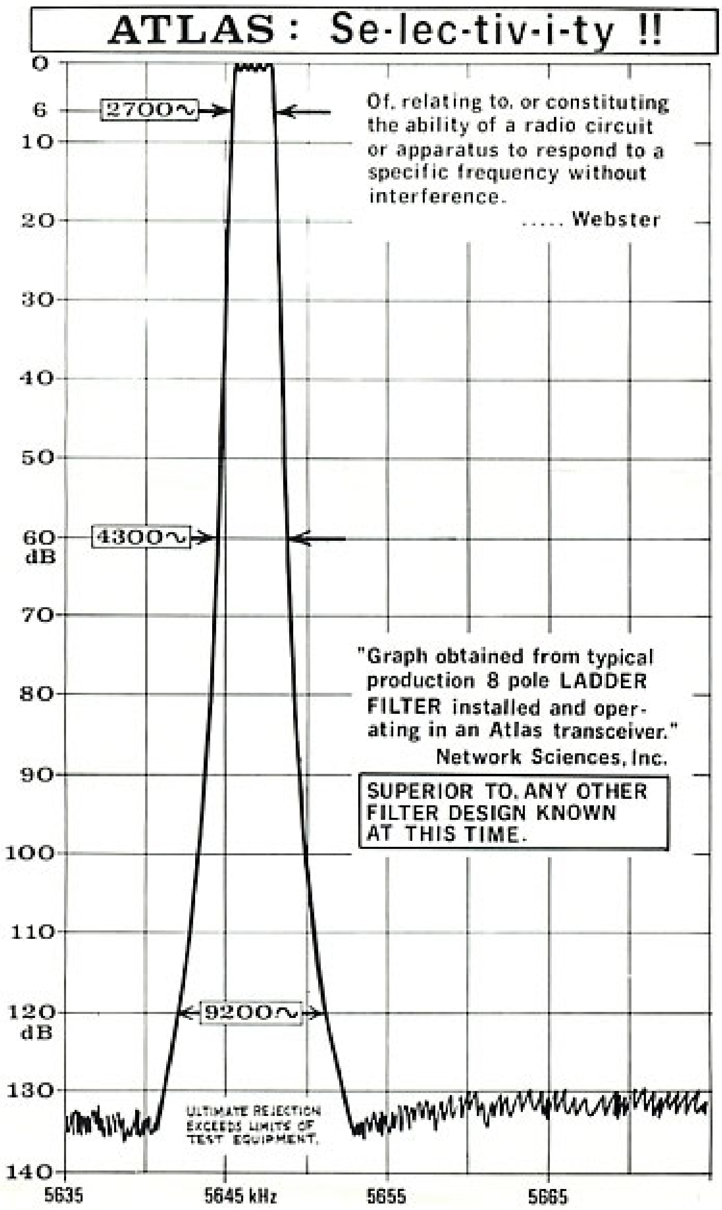
RECEIVER DESIGN. Results in spectacular performance. The very latest state-of-the-art techniques include a double balanced doide mixer, and a minimum of signal amplification before coupling into the I.F. crystal filter. As a result, the receiver's ability to handle very strong adjacent channel signals without overload and intermodulation establishes new standards of comparison. If you have not yet operated an Atlas transceiver in a crowded band and compared it with others, you have a real thrill coming.

MAXIMUM OPERATING PLEASURE. The front panel design of the Atlas has full size knobs and tuning dial, in spite of its small overall size. Small size should not sacrifice operating pleasure. Our fingers and hands do not become smaller. You'll find operating the Atlas transceiver a delightful experience.

MODULAR CONSTRUCTION. Most of the circuitry is on printed circuit boards. There are three plug-in boards for R.F., I.F., and A.F. circuits which provides easy servicing, when required. All sections are readily accessible.

CUSTOMER SERVICE SECOND TO NONE. Built to the best commercial specs. Service will not be required very often. But, when required, Atlas Radio backs its product with a service policy second to none. **Your satisfaction is guaranteed!**

SUPER SELECTIVITY



- **SUPER SELECTIVITY:** The Atlas transceivers feature an 8 pole crystal ladder filter designed especially for Atlas by Network Sciences of Phoenix, Arizona. This filter represents a major breakthrough in filter design with unprecedented skirt selectivity and ultimate rejection. Its superior selectivity has been tailored to take full advantage of the extremely wide range of signal levels that the Atlas front end is capable of handling.
- **6 db BANDWIDTH,** 2700 cycles. Purposely selected to provide audio response from 300 to 3000 cycles in both transmit and receive modes. Scientific studies by independent laboratories, including Bell labs, have proven that transmission and reception of voice frequencies between 300 and 3000 cycles provides a substantial improvement in readability under noisy or weak signal conditions, as compared to narrower bandwidths. At the same time, the improvement in fidelity of voice communication is readily noticeable, and accounts for the constant reports of "broadcast quality" from Atlas transceivers. An interesting note may be added. Other receivers with narrower bandwidth cannot fully appreciate the audio quality of the Atlas transmitter. It takes 2700 cycles of bandwidth to get all of the quality!
- **SKIRT SELECTIVITY.** The 8 pole ladder filter, as measured in the Atlas transceiver, provides a bandwidth at 60 db down of only 4300 cycles (shape factor of 1.6) and a bandwidth of only 9200 cycles at 120 db down! *No other filter that we know of would even list their 120 db Bandwidth.* Note that the Atlas filter is narrower at these levels than other filters, even though the others provide less bandwidth at 6 db.
- **ULTIMATE REJECTION** is in excess of 130 db, greater than the measuring limits of most test equipment.

IT IS THIS EXTREMELY STEEP SKIRT SELECTIVITY, illustrated in the accompanying chart, which rejects strong adjacent channel signals better than any other known receiver. This amazing selectivity, combined with the superior front end circuit design makes the Atlas receiver better than any other.

THE ATLAS RADIO STORY

Our first product, the Atlas 180 was introduced in January of 1974. This all solid state SSB Transceiver covered the 20, 40, 80, and 160 meter bands, ran 180 watts input, was only 3½ inches high by 9½ inches wide, and 9½ inches deep, and weighed just seven pounds. Its state-of-the-art design was taken directly from the AN/URC-87 military manpack radio designed by the solid state genius, Les Earnshaw, and produced by his company Southcom International.

In 1975, the Atlas 180 evolved into the Model 210 which covered 10 through 80 meters, and the 215 which covered 15 through 160 meters. With further refinements, these are now the current models 210x and 215x.

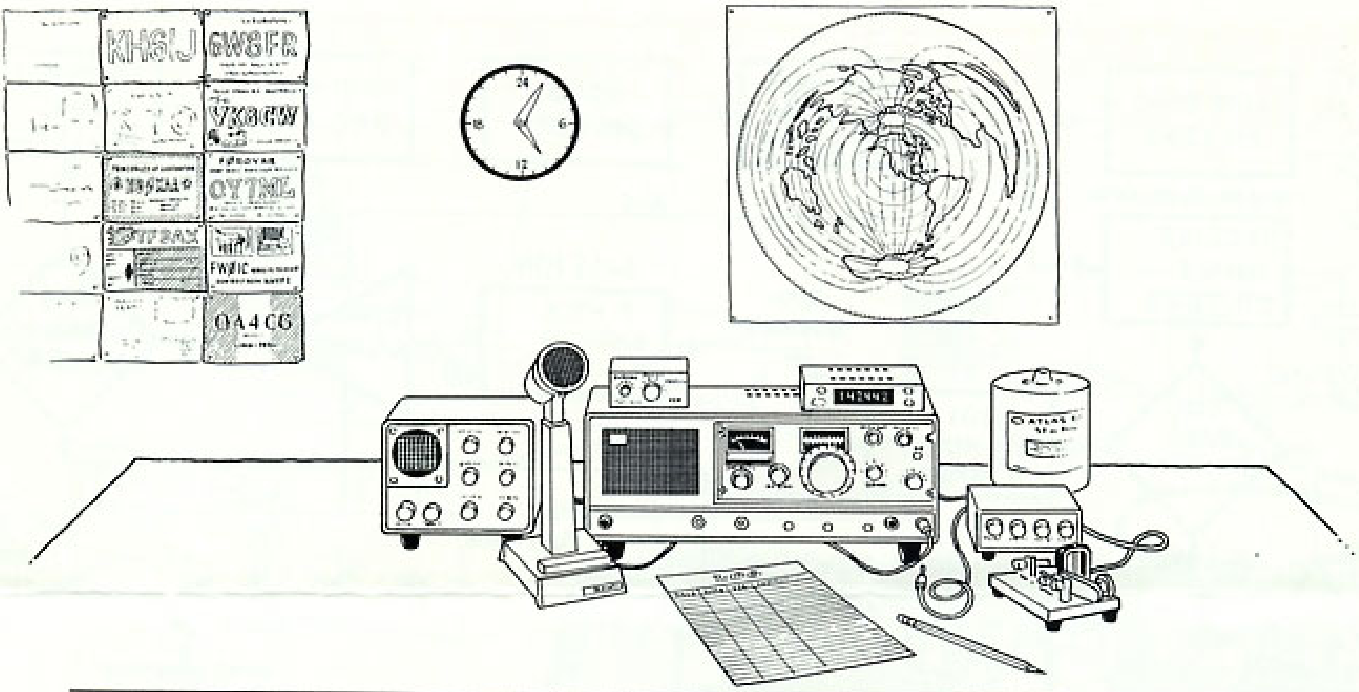
Since January 1974, we have manufactured and sold over 14,000 of these little jewels and their outstanding performance and reliability have become famous all over the world. Consider this: If any service work is required, the radio usually ends up back here at the factory. Weighing only 7 pounds, it's easier to ship it back to us than to pry into its intricate insides. Yet due to the remarkable reliability of the Atlas 210x and 215x, our customer service department needs only two technicians and one secretary in an 18' x 22' room. Turn around time is 2 to 3 days.

Have you noticed how difficult it is to locate an Atlas transceiver on the used market? Check dealer listings on used gear; you'll seldom find an Atlas listed.

MADE IN AMERICA

We're very proud that every Atlas transceiver is made right here in America. We think the American worker, and our employees in particular, are the most talented, industrious people in the world. The quality and versatility of our transceivers are proof of this.

By using American quality workmanship, advanced value engineering in design and manufacture, and rigid quality control, Atlas transceivers are not only competitively priced with the imports, but are actually a better value!



ACCESSORIES

AC CONSOLE, Model 220-CS

Provides for easy plug-in and removal of the 210x or 215x.
Input voltage: Selectable 110 or 220 volts nominal. 50-60 cycle.
Power Consumption: 15 watts, receive mode, 125 watts, average, 250 watt peak, transmit mode. **Output Voltage:** 16 to 13 volts D.C. at 0-16 amps, nominal, and 13 volts regulated D.C. at 1 amp. **Speaker:** 3 x 5 in. 4 ohm, 1.1 oz. magnet. **Accessories:** Space under transceiver provides for addition of VOX and semi-break-in CW. **Finish:** Vinyl cover steel. 2 tone black and grey. **Dimensions:** 15½ in. (39.4 cm) wide, 5⅝ in. (14.3 cm) high, 9½ in. (24.1 cm) deep. **Weight:** 17 lbs. (7.7 Kg) net, 20 lbs. (9.1 Kg) shipping.

PORTABLE AC SUPPLY Model 200-PS 1

Input Voltage: Selectable 110 or 220 volts nominal 50-60 cycle. **Power Consumption:** 15 watts, receive mode. 125 watt average, 250 watts peak, transmit mode. **Output Voltage:** 16 to 13 volts D.C. at 0-16 amps, nominal, and 13 volts regulated at 1 amp. **Connectors:** AC line cord, and D.C. cable with plug for Atlas transceiver. **Dimensions:** 5½ in. (14 cm) wide, 3½ in. (8.9 cm) high, 6½ in. (16.5 cm) deep. **Weight:** 7 bls. 4 oz. (3.3 Kg) net. 10 lbs. (4.5 Kg) shipping.

DC CABLE. 8' long, with reverse polarity protection. Not required if plug-in Mobile Mount is purchased. Includes circuit breaker. Order Model DCC.

2

MT-1. Mobile Antenna matching transformer. Broad-band design transforms base impedance to 50 ohms.

3

DELUXE PLUG-IN MOBILE MOUNT. Provides for easy plug-in and removal of the 210x or 215x. Complete with D.C. Cable, reverse polarity protection, circuit breaker, and Mounting Hardware. Order Model DMK.

MOUNTING BRACKET KIT. Includes Brackets and Hardware for mobile installation. Not required if Plug-in Mobile Mount is purchased. Order Model MBK.

4

NOISE BLANKER. Model PC-120. Operates ahead of the crystal filter. Highly effective on ignition noise, Loran signals, and other pulse type interference. Installs inside Atlas transceiver in place of PC-100 plug-in card. Panel switch for ON and OFF. Transceivers may be ordered with Blanker installed, or can be easily owner or dealer installed.

5

CRYSTAL OSCILLATOR ACCESSORY. Model 10XB. Provides 10 crystal positions. Vernier frequency control. Cable plugs into back of Atlas transceiver. Switch selects crystal or VFO controls. Dimensions: 3½ in. wide, 1½ in. high, 4½ in. deep. Wt: 10 oz.

VOX ACCESSORY. Installs in lower section of AC console. Controls for VOX Gain, anti-VOX, and Delay are brought out to front panel. Also provides for semi-break-in CW operation. AC consoles may be ordered with VOX installed, or can be easily owner or dealer installed. Model VX-5.

6

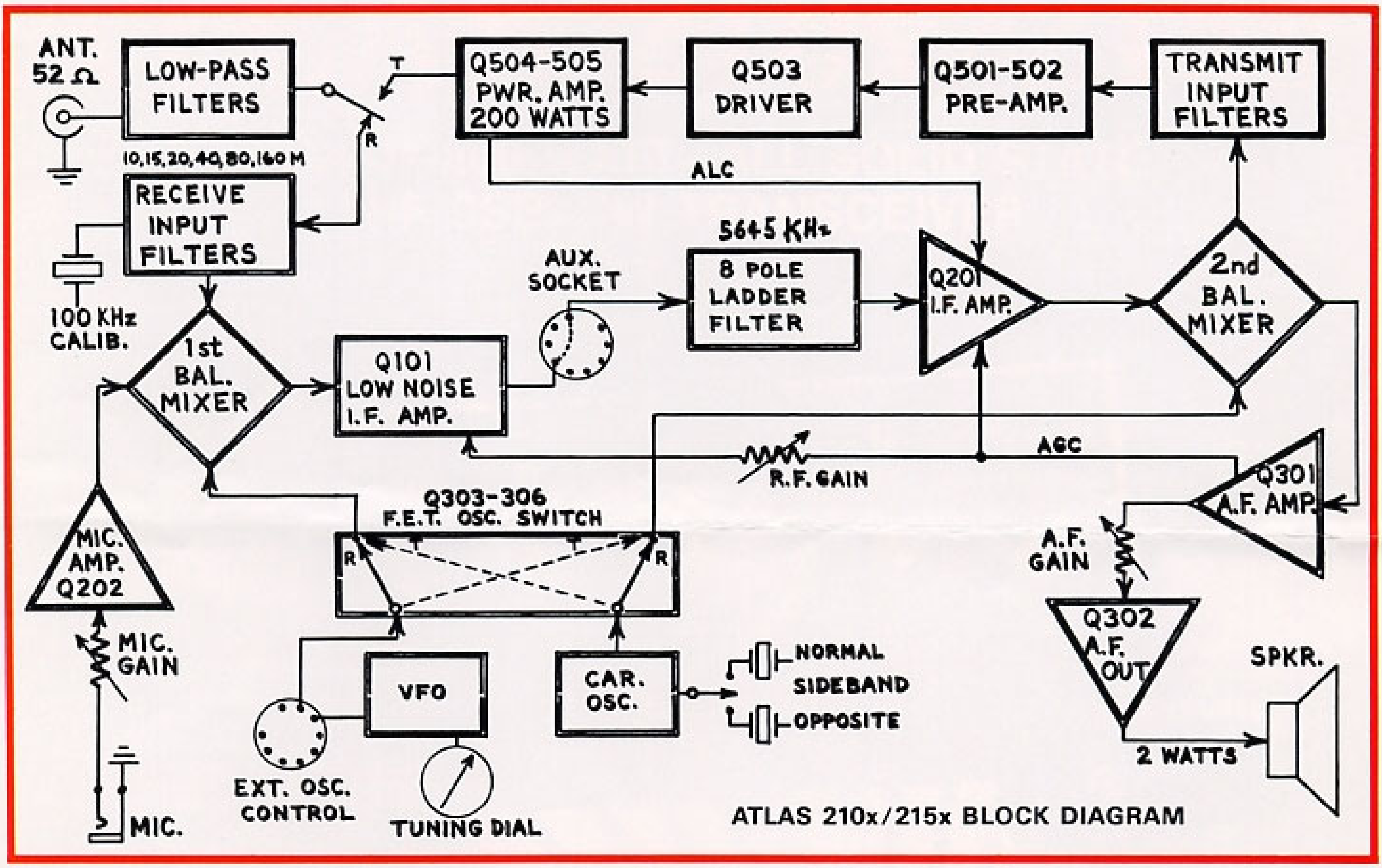
MODEL VX-5M. Same as VX-5 but is self-contained in its own cabinet for mobile or portable use.

7

DUMMY LOAD MODEL DL200 52 OHM. 200 watt intermittent or 60 watt continuous power rating. Housed in a compact one quart can.

8

DD-6 DIGITAL DIAL. With built-in frequency counter capability. All L.E.D. Dot Matrix Display reads frequency to within 50 Hz. Can be used with all Atlas transceivers.



ATLAS 210x/215x BLOCK DIAGRAM

ATLAS 210x/215x SPECIFICATIONS

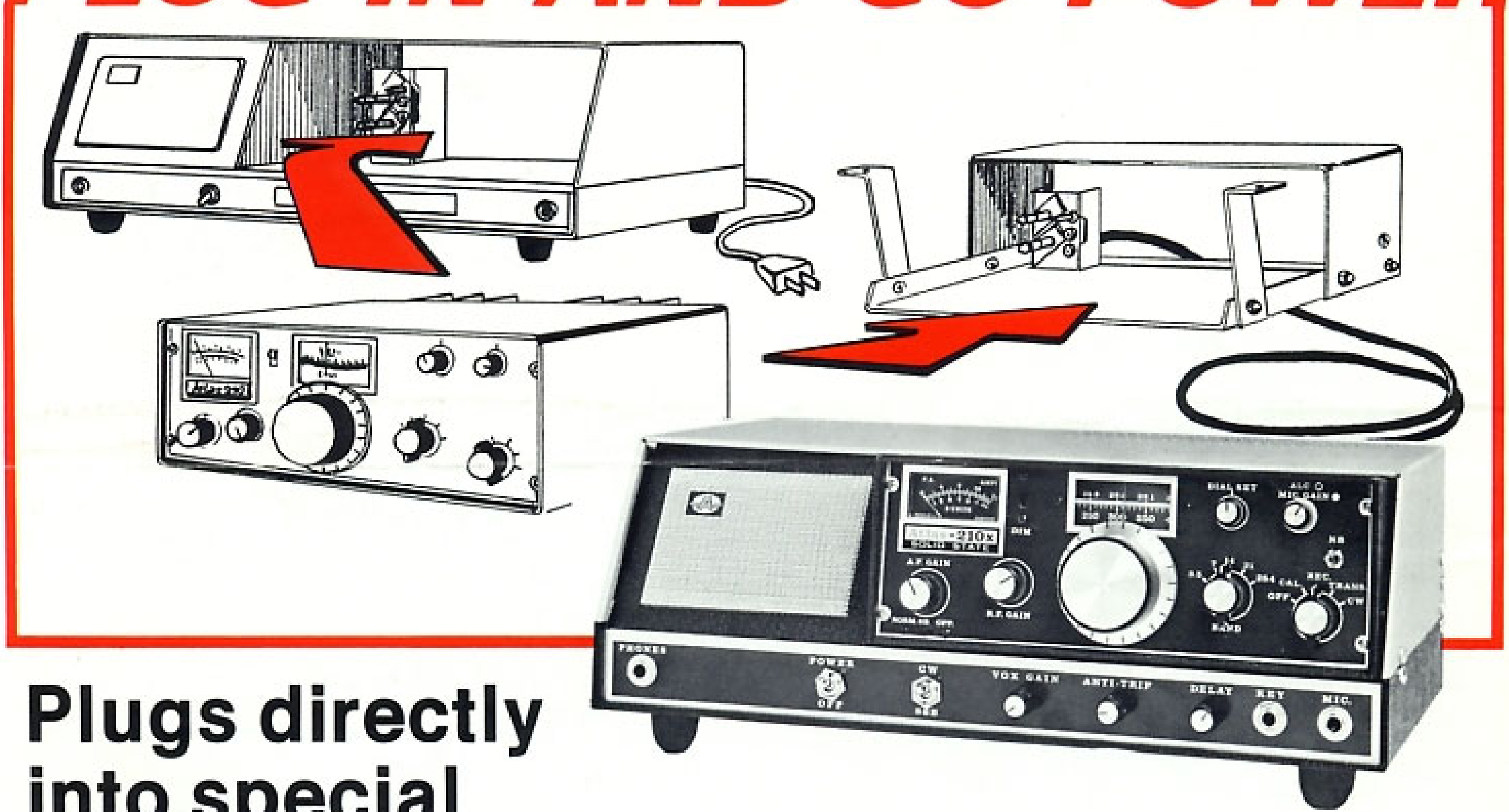
GENERAL: Frequency Coverage with Internal VFO: 1800–2000 kHz, (Model 215x only), 3500–4000 kHz, 7000–7500 kHz, 14,000–14,500 kHz, 21,000–21,500 kHz, 28,400–29,400 kHz, (Model 210x only). **Note** that the 10 meter band may be easily owner adjusted to cover any 1000 kHz segment.

Frequency Control: Highly stable VFO common to both receive and transmit modes. Tuning dial calibrated in 5 kHz increments with 1 kHz increments on skirt of tuning knob, except on 10 meters where increments are 10 kHz and 2 kHz, respectively. Tuning rate is 22 kHz per revolution. **Frequency Stability:** Less than 1 kHz drift during first 30 min. (2 kHz max. on 10 meters). Less than 300 Hz per hour after 30 min. Less than 100 Hz shift from 11 to 14 volts supply. **External Frequency Control:** Rear socket provides for plug-in of external VFO or crystal oscillator for separate control of transmit and receive frequencies, or for network and MARS operation. **Frequency Coverage with Crystal Oscillator:** (1800–3000 kHz, model 215x only), 3000–5200 kHz, 5800–10,000 kHz, 13,800–14,900 kHz, 20,600–21,600 kHz, 28,000–30,000 kHz, (Model 210x only). **Completely Solid State:** Includes 4 I.C.'s, 18 transistors, 32 diodes. **Modes of Operation:** SSB with selectable sideband, and CW. Normal sideband position is LSB on 160, 80, and 40 meters, USB on 20, 15, and 10 meters. Automatic off-set frequency on CW transmit. **Modular Construction:** Plug-in PC boards for R.F., I.F., and audio circuits. **Plug-in Design:** Rear connectors are designed so transceiver plugs into Mobile Mounting Bracket, or AC Console. Connectors are standard: SO-239 coax. antenna jack, 1/4 in. diam. 3 circ. jacks for Mic. and external speaker or headphones, 1/4 in. diam. 2 circ. jack for CW key, 9 pin Noval sockets for Ext. Osc. and Aux. Linear control. **Power Supply Requirements:** 12–14 volts D.C., negative ground only. Terminal P1 is high current circuit for power amplifier, 16 amps. peak in transmit mode. Terminal P2 is low current circuit for receiver and low level stages, draws 300 to 600 ma. in rec. and trans. modes. **Finish:** Vinyl covered aluminum cabinet, black. Anodized aluminum panel. **Dimensions:** 9½ in. (24.1 cm) wide, 3½ in. (8.9 cm) high, 9½ in. (24.1 cm) deep. **Weight:** 6 lbs. 14 oz. (3.1 kg) net. 8 lbs. 6 oz. (3.8 kg) shipping.

RECEIVER SPECIFICATIONS: Circuit Design: Direct conversion of signal to 5645 kHz I.F. using double balanced diode ring mixer, providing exceptional immunity to overload and cross modulation. **Sensitivity:** Requires less than 0.4 microvolts for 10 db signal-plus-noise to noise ratio, 160 through 15 meters. Less than 0.6 microvolts on 10 meters. **Selectivity:** Crystal ladder 8 pole filter. Bandwidth: 2700 Hz at 6 db down, 4300 Hz at 60 db, and only 9200 Hz at 120 db. Ultimate rejection greater than 130 db. 1.6 shape factor. **Image Rejection:** Better than 60 db. **Internal Spurious:** Less than equivalent 2 microvolt signal. **AGC:** Audio output constant within 4 db with signal variation from 5 microvolts to more than 3 volts. **Overall Gain:** Less than 1 microvolt for 0.5 watts audio output. (CW carrier, 1000 Hz heterodyne.) **Audio Output:** 2 watts at 10% distortion, 300 to 3000 Hertz, plus or minus 3 db. **Internal Speaker:** 3 in., 4 ohm, .68 oz. magnet. Rear jack permits plug-in of external speaker or low impedance headphones. AC console automatically disconnects internal speaker and connects front facing speaker. Plug-in Mobile Mount provides for automatic connection of external speaker if desired. **Meter:** Reads "S" units from 1 to 9, plus 10 to 50 db. **Calibrator:** Provides calibration markers at 100 KHz increments on tuning dial. **Dial Set:** Permits adjustment of dial scale calibration.

TRANSMITTER SPECIFICATIONS: Circuit Design: Broadband design eliminates transmitter tuning. Single conversion produces minimum spurious mixing products. 2 section low-pass filters on each band provide excellent harmonic and TVI suppression. ALC with panel adjustment. Infinite SWR protection. **Frequency Control:** Internal VFO automatically transmits exactly on receive frequency. Rear socket provides for plug-in of external VFO or crystal oscillator accessory, (Model 10-XB), for separate control of transmit and receive frequencies, or for network and MARS operation. **Power Rating:** 200 watts P.E.P. input, and CW input, (50 ohm nonreactive load, and **13.6 DC** supply voltage) 160 through 15 meters. 120 watts on 10 meters. **Power Output:** 80 watts minimum P.E.P., and CW on 160 through 15 meters. 50 watts min. on 10 meters. **Note:** Ratings are at 13.6 DC volts to transceiver at full load. **RTTY/SSTV Power Rating:** Approx. 90 watts input, depending on heat sink ventilation. Small fan recommended. **Unwanted Sideband:** More than 60 db down at 1000 Hz audio input. **Carrier Suppression:** More than 50 db down. **Third Order Distortion:** Approx. 30 db below peak power. **Harmonic Output:** More than 40 db below power peak. **CW Transmit:** Manual send-receive. Semi-break-in with CW accessory installed in AC console. Automatic off-set transmit freq. **Transmit Control:** Press-to-talk with Mic. button, or manual transmit with panel switch. Automatic voice control when VOX is installed in AC console. **Microphone:** Dynamic or Crystal, high impedance. Requires 1/4 in. diam. 3 circ. phone plug. **Audio Fidelity:** 300 to 3000 Hz, plus or minus 3 db. **Meter:** Reads P.A. collector current, 0–16 amps. **Linear Amplifier Control:** Aux. socket on rear provides for keying of linear.

PLUG-IN-AND-GO-POWER



Plugs directly into special DC Mobile Mount...or into AC Console

The Deluxe Plug-in Mobile Mount has specially designed rear connectors for DC power input, antenna jack, and mic. jack, that match the same connectors on the Atlas transceiver. Thus all necessary connections for mobile operation are made in seconds by merely sliding the unit into the mobile mount.

Fixed station operation is achieved in the same easy manner, since the Atlas Model 220-CS Console Supply has the same rear connector system as the mobile mount. The internal speaker is automatically disconnected, and the front facing speaker in the console is turned on. Also, the mic. jack is brought out to the front, as well as a headphone jack. The Model 220-CS operates on either 110 or 220 volts, selectable, 50-60 cycles.

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